Nuclear Astrophysics

Exercise 2

Nuclear reaction networks are crucial to understand the origin of elements in the Universe. In many astrophysical events, thousands of nuclei are involved and extended nucleosynthesis networks become necessary to describe the evolution of abundances. In this exercise, we consider a reduced system that can also be solved analytically, consisting only of neutrons, ^{64}Ni, and ^{65}Ni. As part of this exercise you will run the nuclear reaction network code WinNet and compare the final abundances with the analytical solution. For simplicity we assume the system is at a constant temperature of 5 GK and density of $10^9\,\mathrm{g\,cm^{-3}}$ and the initial composition consists entirely of ^{65}Ni. Consider the evolution of the abundances for a time of 10^6 s.

1. Write down which reactions are possible between these three nuclei (n, ^{64}Ni, and ^{65}Ni). How many reactions does this reduced system have?
2. For a reaction network calculation with WinNet, a file with relevant parameters is necessary. Modify the file "test.par" and fill it with the necessary parameters that correspond to the simple model outlined above. Relevant parameters are:
   - *trajectory_mode*
   - *T9_analytic* and *rho_analytic*
   - *read_initial_composition* and *seed_file*
   - *net_source*
   - *termination_criterion* and *final_time*
   - *track_nuclei_every* and *track_nuclei_file*
   - *solver*
   - *reaclib_file*
   - *isotopes_file*

   You can have a look at all parameters here: `https://nuc-astro.github.io/WinNet/parameters.html`
3. Run the reaction network within the Jupyter notebook "runWinNet.ipynb". If something goes wrong, an error message will be given in the file "ERR", otherwise several output files should be created.
4. Look at the file "tracked_nuclei.dat". Plot the abundances of neutrons, ^{64}Ni, and ^{65}Ni by reading and plotting this file with python into a double logarithmic plot. What do you observe?
5. To understand and check the WinNet results, calculate analytically the expected final abundances by writing down the relevant differential equations that are a sum over all reactions:

$$\dot{Y}_i = \underbrace{\sum_j N^i_j \lambda_j Y_j}_{\text{decays and photodisintegrations}} + \underbrace{\sum_{j,k} \frac{N^i_{j,k}}{1+\delta_{jk}} \rho N_\mathrm{A} \langle \sigma v \rangle_{j,k} Y_j Y_k}_{\text{two-body reactions}}, \tag{1}$$

   where we ignored reactions with more than two reactants. Here, N^i_j is the amount of destroyed or created nuclei per reaction, λ is the reaction rate, $\langle \sigma v \rangle$ is the cross section, ρ is the density,

N_A is Avogadro's number, and $\delta_{i,j}$ is a correction factor to prevent double counting: $\delta_{i,j} = 1$ if $i = j$, else $\delta_{i,j} = 0$. Use

$$\lambda_{^{65}\mathrm{Ni}}(5\,\mathrm{GK}) = 2.5778301257 \times 10^{10}\,\mathrm{s}^{-1}\,, \qquad N_\mathrm{A}\langle\sigma v\rangle_{n,^{64}\mathrm{Ni}} = 1.670521 \times 10^{6}\,\mathrm{cm}^3\mathrm{s}^{-1}\mathrm{mol}^{-1}\,. \quad (2)$$

Furthermore, assume that eventually an equilibrium is obtained and make use of mass and charge conservation. Plot the obtained values as horizontal lines into your previous plot. Can you reproduce the values of the simulation?

Hint:

This exercise sheet requires the use of the computational cluster CCSN (unless you managed to install WinNet on your own computer). Connect to your personal Jupyter Notebook with the information given in "General information on the exercises" on Moodle. All necessary files are on Moodle and also already on CCSN.